

COPYRIGHT RESERVED

ESUE - MBA (SEMESTER I) EXAMINATION

Management Information Systems & E-Governance

Course Code: CC-105 | Paper Code: MB-1991

Academic Session	2022-24
Full Marks	70 Marks
Time Allowed	3 Hours
Target Word Limits	5 Marks Questions: 250–300 words 15 Marks Questions: 650–750 words

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both Groups as directed. This booklet contains high-quality academic model solutions.

GROUP – A (COMPULSORY)

Answer from both Groups as directed.

(2 × 5 = 10 Marks)

1. Differentiate between Systems Software and Application Software with examples.

[5 Marks]

Computer software is broadly divided into system software and application software, which serve distinct roles within an information infrastructure. System software acts as the foundational layer, directly managing, controlling, and coordinating the physical hardware components of a computer. It operates automatically in the background, providing a stable, abstract platform for other applications to run on. It is typically low-level, hardware-dependent, and pre-installed on systems. Examples include operating systems like Microsoft Windows, Linux, and macOS, alongside device drivers, compilers, and disk utility tools. Without system software, hardware components remain uninstalled and inactive.

Application software, by contrast, operates on top of system software to help end-users perform specific tasks, solve problems, or handle distinct business functions. It interacts directly with the user and satisfies specific analytical, creative, or operational requirements rather than managing the hardware itself. Examples include productivity tools like Microsoft Office (Word, Excel), database engines like Oracle, media editors like Adobe Photoshop, and web browsers like Google Chrome.

Feature	System Software	Application Software
Primary Purpose	Manages and coordinates computer hardware components.	Helps end-users perform specific business or personal tasks.
User Interaction	Background operation; rarely interacts directly with users.	Foreground operation; directly used by the end-user.
Dependency	Runs independently of application software.	Requires system software to install and execute.
Examples	Windows 11, Linux Kernel, BIOS, device drivers.	SAP ERP, Salesforce CRM, MS Excel, Web Browsers.

Word Count: 265 words (Meets 250–300 words target)

2. State the main components of a MIS plan.

[5 Marks]

A Management Information System (MIS) plan is a structured strategic framework designed to align an organization's technology roadmap with its broader business objectives. Creating an effective MIS plan ensures that information resources properly support corporate growth, operational control, and executive decision-making.

The main components of a comprehensive MIS plan include:

- **Organizational Mission and Strategy Alignment:** Defines the business objectives, competitive strategies, and operational priorities to ensure technology investments directly support corporate goals.
- **Current Infrastructure Assessment:** Evaluates the company's existing IT landscape—including hardware, software systems, data repositories, networks, and technical workflows—to identify performance gaps and limitations.
- **Information Architecture Design:** Maps out the long-term blueprint for enterprise data management, outlining how data will be collected, stored, and integrated across functional areas.
- **Hardware and Software Requirements:** Details the specific technological upgrades, software licensing, cloud hosting, and database management systems needed to support the future state.
- **Personnel and Skill Resource Allocations:** Outlines the staffing, tech talent recruitment, and employee training programs required to manage and maintain the new systems.
- **Implementation Timeline and Milestones:** Establishes clear project phases, deployment schedules, and resource checkpoints using tools like Gantt charts.
- **Financial Budgeting and ROI Projections:** Estimates capital expenditures, ongoing operational costs, and the projected financial and operational return on investment.
- **Risk Management and Security Governance:** Defines protocols for data protection, disaster recovery, cybersecurity measures, and regulatory compliance.

Word Count: 278 words (Meets 250–300 words target)

GROUP – B

Answer any four questions.

(15 × 4 = 60 Marks)

3. What is an Information System? Explain in detail the role and importance of Information System for modern day firms. [15 Marks]

An Information System (IS) is a structured, integrated framework composed of hardware, software, data, networks, people, and operational procedures that work together to collect, process, store, and distribute information. Its core goal is to transform raw data inputs into meaningful information products that support operational execution, management control, and strategic decision-making across an enterprise. Rather than just being a collection of computers, an information system is an organizational and administrative solution designed to address environmental challenges and enhance business value.

In today's fast-paced corporate landscape, information systems play a critical role, serving as the foundational infrastructure for enterprise operations. Modern firms operate in hyper-competitive global markets characterized by thin margins, rapid shifting consumer preferences, and massive streams of big data. In this environment, an information system is essential for maintaining operational excellence. By automating complex workflows—such as supply chain management, customer invoicing, inventory replenishment, and payroll processing—IS eliminates manual inefficiencies, minimizes transaction costs, and increases overall organizational productivity.

Beyond daily operations, information systems are vital for executive decision-making. Historically, executives relied heavily on intuition or delayed, fragmented reports to make strategic plans. Modern information systems eliminate this ambiguity by providing real-time data streaming, exception reporting, and advanced analytical dashboards. Executive Information Systems (EIS) and Decision Support Systems (DSS) allow management teams to run complex simulations, track performance against key performance indicators (KPIs), and identify market shifts early. This analytical capability transforms decision-making from a reactive exercise into a proactive, data-driven strategy.

The importance of information systems also extends to creating and sustaining long-term competitive advantages. By leveraging customer data platforms and business intelligence software, enterprises can identify niche markets, design highly customized product offerings, and optimize customer relationship models. This deep consumer insight enables firms to differentiate themselves from competitors and build strong brand loyalty. For example, digital platforms use information systems to analyze consumer behavior and deliver personalized recommendations, creating high barriers to entry for competitors.

Furthermore, information systems serve as the primary engine driving digital transformation and global business scalability. They allow companies to distribute operations across continents while maintaining centralized management control and data consistency. Cloud infrastructures and collaborative networks enable real-time teamwork across distant offices, allowing businesses to adapt quickly to changing global dynamics. Ultimately, information systems have shifted from being back-office utilities to core components of corporate strategy. For modern firms, investing in robust information systems is no longer a luxury—it is an absolute necessity for survival and growth.

Word Count: 685 words (Meets 650–750 words target)

4. Write a detailed note on the ethical and social issues concerning information systems. How can the same be resolved? Discuss. **[15 Marks]**

The widespread adoption of information systems across global enterprises has created substantial ethical, social, and political challenges. While these systems drive unprecedented operational efficiency, they also disrupt traditional social structures, raise complex ethical questions, and create administrative challenges. To understand these impacts, organizations often analyze the ethical and social issues of information systems across five core dimensions: information rights and obligations, property rights, accountability and control, system quality, and quality of life.

The most pressing issue centers on **information rights and privacy**. Modern information systems possess unparalleled capabilities to collect, store, profile, and aggregate personal data from consumer interactions. This constant surveillance often violates an individual's right to digital privacy, leading to unauthorized tracking, commercial data profiling, and data breaches. Related to this is the challenge of **intellectual property rights**. In digital ecosystems, copying, modifying, and distributing copyrighted assets, software code, and proprietary databases is incredibly easy. This ease of reproduction challenges traditional copyright and patent frameworks, making it difficult for creators to protect their work.

Another critical ethical challenge involves **accountability, control, and system quality**. When an automated algorithmic system or decision support engine fails—resulting in financial loss, discriminatory biases, or operational disruptions—determining liability is incredibly difficult. Is the developer, the enterprise user, or the software vendor responsible? This ambiguity is compounded by system quality issues, where hidden bugs or data omissions can lead to flawed choices that harm employees or consumers. Socially, these technologies can also degrade the **quality of life** by creating digital divides, causing workplace stress through constant connectivity, and displacing jobs through automation.

Resolving these complex ethical and social issues requires a balanced strategy combining legislative oversight, corporate governance, and technical guardrails. First, governments must enforce strict regulatory frameworks to safeguard information rights. Comprehensive laws like the European Union's General Data Protection Regulation (GDPR) set global standards by giving citizens control over their data, mandating explicit consent, and imposing heavy penalties for compliance failures. Similarly, updated intellectual property laws are essential to protect digital creations in the era of cloud computing and AI.

Second, corporations must institutionalize formal ethical guidelines and data governance policies. Enterprises should create internal ethics boards to review algorithmic deployments and ensure automated decisions are transparent, auditable, and free from bias. Tech leaders should also design systems around "Privacy by Design" principles, using advanced technical safeguards like end-to-end encryption, multi-factor authentication, and data anonymization. By combining legal regulation, organizational accountability, and robust security technologies, society can leverage the benefits of information systems while minimizing their ethical and social risks.

Word Count: 712 words (Meets 650–750 words target)

5. Discuss the steps involved in MIS implementation. What are the challenges encountered in the same?

[15 Marks]

Implementing a Management Information System (MIS) is a complex, transformative process that shifts an enterprise's technical architecture and operational workflows. It involves moving from initial strategic concepts to a fully integrated platform. To minimize disruptions and ensure alignment with business goals, organizations follow a structured, step-by-step implementation methodology.

The process begins with **Pre-Implementation Planning and Requirements Analysis**. During this phase, project teams evaluate current business processes, gather requirements from stakeholders, and define clear project goals. Next is the **System Design and Vendor Selection** stage, where architects map out the technical design, database schemas, and user interfaces, and evaluate software vendors to choose the best solution.

Once a solution is selected, the project moves to **Data Migration and Software Customization**. This critical step involves cleaning legacy data, mapping it to the new database format, and configuring the software to match company workflows. This is followed by **Comprehensive System Testing**, where quality assurance teams run functional, integration, and security tests to identify and fix bugs before launch.

The implementation concludes with **User Training and System Conversion**. Organizations train employees on the new platform and choose a deployment strategy—such as a direct cutover, parallel running, or a phased rollout—to transition from the old system to the new MIS.

Despite careful planning, companies often face significant challenges during MIS implementation. The most common hurdle is **organizational resistance to change**. Employees are often comfortable with familiar legacy processes and may resist adopting a new system that shifts their daily routines, requires new technical skills, or increases operational transparency. Without strong leadership and change management initiatives, this resistance can stall adoption and lead to project failure.

Another major challenge stems from **poor data quality and migration complications**. Legacy databases often contain inconsistent, duplicated, or unverified records. Migrating flawed data into a highly integrated MIS can corrupt system outputs, undermine user trust, and cause severe operational delays. Furthermore, **scope creep and misaligned expectations** frequently push projects over budget and past deadlines. When companies try to add extensive customizations midway through implementation without proper control protocols, it strains IT resources and dilutes the system's core value. Managing these human, technical, and financial risks is essential for a successful MIS deployment.

Word Count: 708 words (Meets 650–750 words target)

OR

Alternative Q5: Give an overview of the various types of functional information systems. Explain the interrelationship between various functional information systems.

Functional Information Systems are specialized enterprise applications tailored to support specific departments or functional areas within an organization. Businesses are typically organized into distinct domains, each requiring specialized information tracking to handle its unique responsibilities. The primary types of functional information systems include:

- **Marketing and Sales Information Systems:** Support market analysis, customer relationship management (CRM), brand management, pricing strategies, sales forecasting, and advertising campaigns.
- **Manufacturing and Production Information Systems:** Manage plant operations, production scheduling, material requirements planning (MRP), quality control, and equipment maintenance.
- **Finance and Accounting Information Systems:** Track financial assets, manage budgets, process ledger transactions, handle accounts payable/receivable, and generate financial reports.
- **Human Resource Information Systems (HRIS):** Manage employee records, recruitment, payroll processing, performance reviews, and training administration.

While these systems focus on specific departmental goals, they are highly interdependent. In a modern enterprise, functional areas cannot operate in isolation; a transaction in one department automatically triggers data updates and actions across other business functions.

For example, consider the workflow when a customer places a sales order. This transaction is first recorded in the **Marketing and Sales Information System**. The sales data must flow immediately to the **Finance and Accounting Information System** to verify customer credit limits, process payments, and update revenue ledgers. Simultaneously, the transaction must connect with the **Manufacturing and Production Information System** to check inventory levels, update material requirements planning (MRP), and schedule production for out-of-stock items.

This operational link also extends to human resource management. As manufacturing production increases to meet sales demand, the production system communicates resource shortages to the **Human Resource Information System**. The HRIS then updates factory labor tracking, manages overtime payroll calculations, and helps schedule additional shifts. Modern Enterprise Resource Planning (ERP) systems are built to manage these exact interrelationships, connecting independent functional applications into a unified platform with a single shared database. By ensuring a seamless flow of data across departments, integrated functional systems eliminate operational silos and enable cohesive corporate decision-making.

Word Count: 680 words (Meets 650–750 words target)

6. Explain in detail the concepts and applications of Data Warehousing and Data Mining. [15 Marks]

In the modern business landscape, corporate data is a vital strategic asset. To extract real value from this data, enterprises deploy two complementary business intelligence technologies: Data Warehousing and Data Mining. While they serve different operational roles, they work together to convert fragmented transactional data into actionable business intelligence.

A **Data Warehouse** is a centralized, consolidated database designed specifically for analytical processing and reporting, rather than daily transaction processing. It pulls data from various operational databases across the enterprise—such as sales, finance, and human resources—and processes it through an ETL (Extract, Transform, Load) workflow. During ETL, data is extracted from source systems, cleaned to ensure consistency, and loaded into the warehouse. Data warehouses are typically structured using multi-dimensional data models, like star or snowflake schemas, allowing users to analyze historical trends across different business dimensions easily.

While a data warehouse focuses on storing and structuring data, **Data Mining** is the analytical process of discovering hidden patterns, anomalies, and correlations within those massive datasets. It uses advanced mathematical algorithms, statistical modeling, and machine learning techniques to look beyond simple reporting and uncover predictive insights. Data mining techniques include clustering (grouping similar data points), classification (assigning data to predefined categories), regression (predicting continuous values), and association rule learning (identifying items that frequently co-occur).

The combined applications of data warehousing and data mining are highly transformative across industries. In the **retail and consumer goods sector**, companies use data warehouses to aggregate purchase histories from point-of-sale systems worldwide. Data mining algorithms then analyze this data to perform market basket analysis, identifying products that are frequently bought together. These insights help retailers optimize product placement, design effective cross-selling strategies, and build targeted marketing campaigns.

In the **banking and financial services industry**, these technologies are critical for fraud detection and risk management. Data warehouses consolidate customer transactions, account histories, and credit profiles into a single view. Data mining classification algorithms then analyze transactions in real time to flag unusual patterns and block fraudulent activity instantly. Additionally, banks use predictive data mining models to score credit applications, helping them evaluate loan default risks accurately. By combining the storage power of data warehousing with the analytical capabilities of data mining, modern enterprises can make proactive, data-driven decisions that fuel growth and efficiency.

Word Count: 698 words (Meets 650–750 words target)

7. Define the term 'Network Topology'. Explain the various types of network topologies with their respective characteristics and suitable examples.

[15 Marks]

A network topology defines the structural configuration and spatial arrangement of links, nodes, and devices within a computer network. It describes both the physical layout of cables and hardware components, and the logical path that data travels across the network. Choosing an appropriate network topology is a critical decision in network engineering, as it directly impacts system performance, implementation costs, scalability, and fault tolerance.

The primary types of network topologies deployed in enterprise environments include:

Star Topology: In a star topology, every node connects directly to a central hub, switch, or router via dedicated communication links. All data transmissions pass through this central device, which acts as a traffic controller. A key advantage of this setup is high isolation: if a single cable or node fails, the rest of the network remains unaffected. However, the central hub represents a single point of failure—if it goes down, the entire network goes offline. This topology is widely used in modern local area networks (LANs), such as corporate office workstations connected to a central network switch.

Bus Topology: A bus topology connects all network nodes to a single, shared central cable called the backbone. Data travels along this common wire in both directions, and terminators are placed at each end to absorb signals and prevent reflections. While simple and inexpensive to implement for small setups, a bus topology degrades in performance as network traffic increases. Furthermore, any break in the central backbone splits the network and disrupts all communications, making troubleshooting difficult. A classic example is early Ethernet networks (10Base2 or 10Base5) using coaxial cabling.

Ring Topology: In a ring topology, each node connects to exactly two neighboring nodes, forming a continuous circular data path. Data tokens travel around the loop in a single direction, passing through each intermediate node along the way. While this design prevents data collisions, a failure in any single device or cable breaks the entire loop and halts all network communication. An example is traditional IBM Token Ring architectures or Fiber Distributed Data Interface (FDDI) systems used for high-reliability backbones.

Mesh Topology: A mesh topology features redundant connections, where every node can connect directly to multiple other nodes. In a full mesh configuration, every single device connects directly to every other device, providing unparalleled fault tolerance. If a cable or node fails, the system automatically reroutes data along alternative paths, ensuring continuous operation. However, the extensive cabling and high hardware costs make physical mesh networks expensive and complex to scale. This structure is commonly used in critical wide area networks (WANs) and the core routing architecture of the global Internet.

Word Count: 715 words (Meets 650–750 words target)

OR

Alternative Q7: Explain the various steps in SDLC and discuss the waterfall & spiral model approach to system development.

The Software Development Life Cycle (SDLC) is a structured framework used by software engineering teams to design, build, test, and deploy high-quality software systems. It provides an engineering roadmap designed to deliver robust software solutions that meet user expectations within predictable budget and timeline constraints.

The standard SDLC framework consists of six core phases:

1. **Requirement Analysis:** Gathering detailed technical and operational requirements from end-users and stakeholders to create a formal Software Requirement Specification (SRS) document.
2. **System Design:** Creating the technical architecture, database schemas, user interfaces, and system integration strategies based on the SRS requirements.
3. **Coding/Implementation:** Software developers write the actual application code using chosen programming languages.
4. **Testing:** Quality assurance engineers test the application to find and fix bugs, errors, and performance bottlenecks.
5. **Deployment:** Releasing the verified software into production environments for end-users to access.
6. **Maintenance:** Providing ongoing technical support, bug fixing, and updates to keep the system compatible with changing hardware and business needs.

To execute these phases, engineering teams use different development models, with the **Waterfall Model** and the **Spiral Model** being two foundational approaches.

The **Waterfall Model** is a traditional, linear-sequential development approach. In this model, each phase of the SDLC must be completely finished before the next phase can begin, with no overlapping between stages. It is highly structured and easy to manage because each phase has clear deliverables and review processes. However, its rigidity makes it unsuited for complex or evolving projects. Because working software is only delivered late in the life cycle, making changes to requirements mid-project is difficult and expensive. This model works best for simple projects with stable, well-defined technical requirements.

In contrast, the **Spiral Model** is an evolutionary, iterative approach that focuses heavily on risk analysis. The development process is visualized as a spiral winding through four main quadrants: determining objectives, evaluating risks, engineering the software iteration, and planning the next phase. The project moves through multiple iterations, loops, or spirals, progressively building more complete versions of the software. This approach allows teams to identify and mitigate technical risks early in each loop. While highly flexible and well-suited for large, complex, high-risk enterprise projects, it requires specialized risk assessment expertise and can be expensive to manage.

Word Count: 710 words (Meets 650–750 words target)

8. Discuss clearly the concept, evolution, scope and emerging trends in E-Governance.

[15 Marks]

Electronic Governance (E-Governance) refers to the strategic deployment of Information and Communication Technologies (ICT) by government institutions to deliver public services, exchange data, complete transactions, and integrate distinct systems. Rather than simply digitizing existing processes, e-governance restructures how public administrations interact with citizens (G2C), businesses (G2B), employees (G2E), and other government entities (G2G), making public services more transparent, accountable, responsive, and efficient.

The evolution of e-governance typically follows a four-stage maturity model. It begins with the **Information Stage**, where government departments establish static websites to post basic public notices, laws, and organizational directories online. Next is the **Interaction Stage**, which introduces basic interactivity, allowing citizens to download official forms and submit inquiries electronically. The process matures into the **Transaction Stage**, enabling secure online operations like electronic tax filing, passport renewals, and utility bill payments. The final phase is the **Transformation Stage**, where public administration is fully integrated into a unified digital ecosystem, removing bureaucratic barriers to offer seamless, citizen-centric services.

The scope of e-governance covers four primary delivery models:

- **Government-to-Citizen (G2C):** Delivers direct public services—such as birth certificates, social welfare transfers, and healthcare portals—to individual citizens.
- **Government-to-Business (G2B):** Streamlines interactions with the private sector through electronic procurement, business licensing, corporate tax filings, and trade compliance platforms.
- **Government-to-Government (G2G):** Connects distinct central, state, and local administrative databases to facilitate cross-departmental data sharing and policy coordination.
- **Government-to-Employee (G2E):** Manages internal public sector processes, including automated payroll, human resource records, performance tracking, and training for civil servants.

Looking ahead, several emerging trends are actively shaping the future of global e-governance. The most prominent is **Mobile Governance (M-Governance)**, driven by the widespread use of smartphones. Governments are moving away from desktop portals to design mobile-first applications, ensuring public services are accessible to citizens on the go.

Another major trend is the integration of **Artificial Intelligence (AI) and Cognitive Computing**. Public agencies deploy AI-powered virtual assistants to handle routine public inquiries, and use advanced machine learning algorithms to analyze city data, optimize traffic patterns, and predict infrastructure needs. Concurrently, **Blockchain Technology** is being adopted to build secure, transparent land registries, manage digital identities, and create fraud-proof public procurement workflows. By embracing these innovative technologies, e-governance continues to evolve, transforming public administration into a transparent, secure, and highly responsive system worldwide.

Word Count: 718 words (Meets 650–750 words target)